

```
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
```

```
#define MAXPAROLA 30
#define MAXRIGA 80
```

```
int main(int argc, char *argv[])
{
    int freq[MAXPAROLA]; /* vettore di contatori
delle frequenze delle lunghezze delle parole */
    char riga[MAXRIGA];
    int i, inizio, lunghezza;
    FILE *f;
```

```
for(i=0; i<MAXPAROLA; i++)
    freq[i]=0;
```

```
if(argc != 2)
```

```
{
    fprintf(stderr, "ERRORE, serve un parametro con il nome del file\n");
    exit(1);
}
```

```
f = fopen(argv[1], "r");
if(f==NULL)
```

```
{
    fprintf(stderr, "ERRORE, impossibile aprire il file %s\n", argv[1]);
    exit(1);
}
```

```
while( fgets( riga, MAXRIGA, f ) != NULL )
```



Synchronization

Introduction to Synchronization

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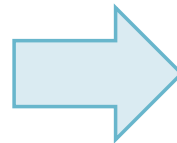
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Introduction

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Introduction

- ❖ Critical Section (**CS**) or Critical Region (**CR**)
 - A section of code, common to multiple threads, in which each thread can read and **write** shared objects
- ❖ Access to CS is subject to **race conditions**
 - The result depends on the execution order of the processes instructions

Example

FIFO, Queue, Circular Buffer

T_i

```
void enqueue (int val) {  
    if (n>SIZE) return;  
    queue[tail] = val;  
    tail=(tail+1)%SIZE;  
    n++;  
    return;  
}
```

register = n
register = register + 1
n = register

T_j

```
int dequeue (int *val) {  
    if (n<=0) return;  
    *val=queue[head];  
    head=(head+1)%SIZE;  
    n--;  
    return;  
}
```

register = n
register = register - 1
n = register

- ❖ Even if **enqueue** and **dequeue** operate on the different ends of the queue, the variable **n** is shared

Race condition:
Increments and decrements can
be lost

Critical sections

- ❖ Race conditions could be prevented with an **access protocol** that enforces **mutual exclusion** for each CS
 - Before a CS, there should be a **reservation** section
 - The reservation code must block (lock out) the P (or T) if another P (or T) is using its CS
 - After the CS, there should be a **release** section
 - The release possibly unlocks another P (or T) which was waiting in the "reservation" code of its CS

Access protocol

T_i

```
while (TRUE) {  
    ...  
    reservation code  
    Critical Section  
    release code  
    ...  
    non critical section  
}
```

T_j

```
while (TRUE) {  
    ...  
    reservation code  
    Critical Section  
    release code  
    ...  
    non critical section  
}
```

- ❖ Every Critical Section is protected by an
 - Enter code (reservation, or prologue)
 - Exit code (release, or epilogue)
- ❖ Non-critical sections should not be protected
- ❖ OSs provide appropriate primitives